Question 1: List the top 3 playlists with the highest average User_Rating and a total Play_Duration_Seconds greater than 5000? The average User_Rating should be rounded to two decimal places

```
SELECT
 Playlist_Name, ROUND(AVG(User_Rating),2) as avg_user_rating ,
SUM(Play_Duration_Seconds) as total_Play_Duration_Seconds
FROM
 `farmers_market.spotify`
GROUP BY Playlist_Name
HAVING  SUM(Play_Duration_Seconds) > 5000
ORDER BY AVG(User_Rating) DESC
LIMIT 3
```

Question 2: List playlists that contain at least one song with a User_Rating of 4.5 or higher and at least one song with a Repeat_Count of 10 or higher.

Columns you need to display: Playlist_Name, user_rating, Repeat_count

```
Select
Playlist_Name, user_rating, Repeat_count
FROM
`farmers_market.spotify`
GROUP BY Playlist_Name, user_rating, Repeat_count
HAVING MAX(user_rating) >= 4.5 and MAX(Repeat_count) >=10
```

```
-- SELECT
-- DISTINCT Playlist_Name,user_rating, Repeat_count
-- FROM `farmers_market.spotify`

-- WHERE Playlist_Name IN (SELECT Playlist_Name FROM `farmers_market.spotify` WHERE User_Rating >= 4.5)
```

```
-- AND Playlist_Name IN (SELECT Playlist_Name FROM `farmers_market.spotify` WHERE
Repeat_Count >= 10);


-- SELECT
-- Playlist_Name,User_Rating,Repeat_Count
-- FROM `farmers_market.spotify`
-- where User_Rating >= 4.5 And Repeat_Count >= 10
-- Group by Playlist_Name, User_Rating, Repeat_Count
-- order by User_Rating desc,Repeat_Count desc;



## Question 3: What are the track names and genres that have been played offline,
exclusively in the state of Delhi and have a total offline play count of at least 5
from Artist Anirudh Ravichander? These track names should also display their total
play count in Delhi, sorted from highest to lowest.

SELECT
   Track_Name,
   Genre,
   SUM(Offline_Plays_Count) AS Delhi_Plays_Count
FROM
   `farmers_market.spotify`
WHERE
   STATE = 'Delhi' and Artist = "Anirudh Ravichander"
GROUP BY
   Track_Name,
   Genre
HAVING SUM(Offline_Plays_Count)>=5
ORDER BY SUM(Offline_Plays_Count) desc;
```

Question 4: In the state of Tamil Nadu, find the tracks by artist Sai Abhyyankar that have been played by either Free or Premium users, but where no user (neither Free nor Premium) has liked the track. Show the Track Name and Track ID.

Hint: Use CTE

```
-- TN and Sai and Sub_type(Free/prem) and track not present in some list

WITH UserLikedSongs AS
(
 SELECT
       DISTINCT Track_Name
   FROM
       `farmers_market.spotify`
   WHERE
       (Subscription_Type = 'Free' AND Liked = True) OR (Subscription_Type = 'Premium'
AND Liked = True)
)

Select
   Track_Name, Track_id
FROM
   `farmers_market.spotify`
WHERE
   State = "Tamilnadu"
   AND Artist = "Sai Abhyyankar"
   AND Subscription_Type IN ("Free", "Premium")
   AND Track_Name NOT IN (Select * from UserLikedSongs)
```

Question 5: List all tracks released after '2025-07-01' of English Language that belong to the top 5 most-listened-to genres, based on total Play_Duration_Seconds. The results should display the Track_Name, Release_Date, Language and the Genre in all capital letters, and be sorted in ascending order by the release date.

Visualize the output
Breakdown the problem

OUTPUT: Track_Name, Release_Date, Language, Genre

released_date > '2025-07-01' and language = English and top5 genre by total(listing hours)

```
WITH top_genre as
(
 SELECT Genre
 FROM `farmers_market.spotify`
 WHERE Release_Date > '2025-07-01'
 GROUP BY genre
 ORDER BY SUM(Play_Duration_Seconds) DESC
 LIMIT 5
)

Select Track_Name, Release_Date, language, UPPER(Genre) as Genre
from `farmers_market.spotify`
WHERE Release_Date > '2025-07-01' and language = "English" and Genre IN (Select * from top_genre)
ORDER BY Release_Date
```

Question 6: Find the users who have a higher average Track_Skip_Count than any other user from their same state. Display the User_Name, Email, User_Age, and Subscription_Type for these users, along with their state

```sql
WITH UserAvgSkips AS
(
 SELECT
   User_Name,
   Email,
   User_Age,
   Subscription_Type,
   State,
   AVG(Track_Skip_Count) as avg_skip_count
 FROM
  `farmers_market.spotify`
 GROUP BY User_Name,Email,User_Age,Subscription_Type,State
)

Select
   User_Name,
   Email,
   User_Age,
   Subscription_Type,
   STATE

from UserAvgSkips as uas
WHERE uas.avg_skip_count > (Select MIN(avg_skip_count) FROM UserAvgSkips where STATE = uas.STATE)

ORDER BY avg_skip_count DESC
```

Question 1: List the top 3 playlists with the highest average User_Rating and a total Play_Duration_Seconds greater than 5000? The average User_Rating should be rounded to two decimal places

```
SELECT
Playlist_Name, ROUND(AVG(User_Rating),2) as avg_user_rating ,
SUM(Play_Duration_Seconds) as total_Play_Duration_Seconds
FROM
`farmers_market.spotify`
GROUP BY Playlist_Name
HAVING  SUM(Play_Duration_Seconds) > 5000
ORDER BY AVG(User_Rating) DESC
LIMIT 3
```

Question 2: List playlists that contain at least one song with a User_Rating of 4.5 or higher and at least one song with a Repeat_Count of 10 or higher.

Columns you need to display: Playlist_Name, user_rating, Repeat_count

```
Select
Playlist_Name, user_rating, Repeat_count
FROM
`farmers_market.spotify`
GROUP BY Playlist_Name, user_rating, Repeat_count
HAVING MAX(user_rating) >= 4.5 and MAX(Repeat_count) >=10
```

```
-- SELECT
-- DISTINCT Playlist_Name,user_rating, Repeat_count
-- FROM `farmers_market.spotify`


-- WHERE Playlist_Name IN (SELECT Playlist_Name FROM `farmers_market.spotify` WHERE
User_Rating >= 4.5)
-- AND Playlist_Name IN (SELECT Playlist_Name FROM `farmers_market.spotify` WHERE
Repeat_Count >= 10);




-- SELECT
-- Playlist_Name,User_Rating,Repeat_Count
-- FROM `farmers_market.spotify`
-- where User_Rating >= 4.5 And Repeat_Count >= 10
-- Group by Playlist_Name, User_Rating, Repeat_Count
-- order by User_Rating desc,Repeat_Count desc;




## Question 3: What are the track names and genres that have been played offline,
exclusively in the state of Delhi and have a total offline play count of at least 5
from Artist Anirudh Ravichander? These track names should also display their total
play count in Delhi, sorted from highest to lowest.


SELECT
  Track_Name,
  Genre,
```

```
  SUM(Offline_Plays_Count) AS Delhi_Plays_Count
FROM
  `farmers_market.spotify`
WHERE
  STATE = 'Delhi' and Artist = "Anirudh Ravichander"
GROUP BY
  Track_Name,
  Genre
HAVING SUM(Offline_Plays_Count)>=5
ORDER BY SUM(Offline_Plays_Count) desc;
```

```
## Question 4: In the state of Tamil Nadu, find the tracks by artist Sai Abhyyankar that have been played by either Free or Premium users, but where no user (neither Free nor Premium) has liked the track. Show the Track Name and Track ID.

## Hint: Use CTE

-- TN and Sai and Sub_type(Free/prem) and track not present in some list

WITH UserLikedSongs AS
(
SELECT
```

```
      DISTINCT Track_Name
  FROM
      `farmers_market.spotify`
  WHERE
      (Subscription_Type = 'Free' AND Liked = True) OR (Subscription_Type = 'Premium'
AND Liked = True)
)


Select
  Track_Name, Track_id
FROM
  `farmers_market.spotify`
WHERE
  State = "Tamilnadu"
  AND Artist = "Sai Abhyyankar"
  AND Subscription_Type IN ("Free", "Premium")
  AND Track_Name NOT IN (Select * from UserLikedSongs)
```

Question 5: List all tracks released after '2025-07-01' of English Language that belong to the top 5 most-listened-to genres, based on total Play_Duration_Seconds. The results should display the Track_Name, Release_Date, Language and the Genre in all capital letters, and be sorted in ascending order by the release date.

```
## Visualize the output
## Breakdown the problem

## OUTPUT: Track_Name, Release_Date, Language, Genre

## released_date > '2025-07-01' and language = English and top5 genre by total(listing hours)



WITH top_genre as
(
SELECT Genre
FROM `farmers_market.spotify`
WHERE Release_Date > '2025-07-01'
GROUP BY genre
ORDER BY SUM(Play_Duration_Seconds) DESC
LIMIT 5
)

Select Track_Name, Release_Date, language, UPPER(Genre) as Genre
from `farmers_market.spotify`
WHERE Release_Date > '2025-07-01' and language = "English" and Genre IN (Select * from top_genre)
ORDER BY Release_Date
```

Question 6: Find the users who have a higher average Track_Skip_Count than any other user from their same state. Display the User_Name, Email, User_Age, and Subscription_Type for these users, along with their state

```
WITH UserAvgSkips AS
(
SELECT
  User_Name,
  Email,
  User_Age,
  Subscription_Type,
  State,
  AVG(Track_Skip_Count) as avg_skip_count
FROM
 `farmers_market.spotify`
GROUP BY User_Name,Email,User_Age,Subscription_Type,State
)

Select
  User_Name,
  Email,
  User_Age,
  Subscription_Type,
  STATE
```

```
from UserAvgSkips as uas
WHERE uas.avg_skip_count > (Select MIN(avg_skip_count) FROM UserAvgSkips where STATE =
uas.STATE)

ORDER BY avg_skip_count DESC
```

Question 7: Identify users who have skipped more songs than they have liked, but only among those who have listened to a minimum of 10 unique tracks.

```
WITH UserEngagement AS (
    SELECT
        User_Name,
        SUM(CASE WHEN Liked = 'Yes' THEN 1 ELSE 0 END) AS Liked_Songs,
        SUM(CASE WHEN Track_Skip_Count > 0 THEN 1 ELSE 0 END) AS Skipped_Songs,
        COUNT(DISTINCT Track_Name) AS Total_Tracks_Listened
    FROM
        spotify
    GROUP BY
        User_Name
)
SELECT
    User_Name,
    Liked_Songs,
    Skipped_Songs,
    Total_Tracks_Listened
FROM
    UserEngagement
```

```
WHERE
    Liked_Songs < Skipped_Songs
    AND Total_Tracks_Listened >=10;
```

Question 8: For genres containing more than 10 tracks, what is the percentage of tracks with 'High' audio quality?. Display the Genre, total tracks, the number of high-quality tracks, and the percentage rounded to two decimal places.

```
WITH GenreCounts AS (
    SELECT
        Genre,
        COUNT(Track_Name) AS Total_Tracks,
        SUM(CASE WHEN Audio_Quality = 'High' THEN 1 ELSE 0 END) AS High_Quality_Tracks
    FROM
        spotify
    GROUP BY
        Genre
)
SELECT
    Genre,
    Total_Tracks,
    High_Quality_Tracks,
    ROUND((High_Quality_Tracks * 100.0) / Total_Tracks, 2) AS High_Quality_Percentage
FROM
    GenreCounts
WHERE
    Total_Tracks > 10;
```

Question 9: Among artists who have released at least 10 tracks with an average user rating above 3, which are the artists with the highest total offline play count? For each of these artists, display their name, total offline plays, total tracks, and their average user rating.

```
WITH ArtistMetrics AS (
```

```
    SELECT
        Artist,
        COUNT(Track_Name) AS Total_Tracks,
        AVG(User_Rating) AS Average_User_Rating,
        SUM(Offline_Plays_Count) AS Total_Offline_Plays
    FROM
        spotify
    GROUP BY
        Artist
)
SELECT
    Artist,
    Total_Offline_Plays,
    Total_Tracks,
    ROUND(Average_User_Rating, 2) AS Average_User_Rating
FROM
    ArtistMetrics
WHERE
    Total_Tracks >= 10
    AND Average_User_Rating > 3
ORDER BY
    Total_Offline_Plays DESC;
```

Question 10: Which users with a 'Free' subscription have:

- Total offline plays greater than 30
- The average user rating higher than the average rating of users from their state?

Show their name, state, total offline plays, and average rating, sorted by total offline plays.

```
WITH UserStats AS (
```

```
    SELECT
        User_Name,
        STATE,
        Subscription_Type,
        SUM(Offline_Plays_Count) AS Total_Offline_Plays,
        AVG(User_Rating) AS User_Avg_Rating
    FROM
        spotify
    GROUP BY
        User_Name,
        STATE,
        Subscription_Type
    HAVING
        SUM(Offline_Plays_Count) > 30
)
SELECT
    User_Name,
    STATE,
    Total_Offline_Plays,
    ROUND(User_Avg_Rating, 2) AS User_Avg_Rating
FROM
    UserStats AS us
WHERE
    us.Subscription_Type = 'Free'
    AND us.User_Avg_Rating > (SELECT AVG(User_Rating) FROM
spotify WHERE STATE = us.STATE)
ORDER BY
    us.Total_Offline_Plays DESC
```